

The CMS Computing Submission Infrastructure

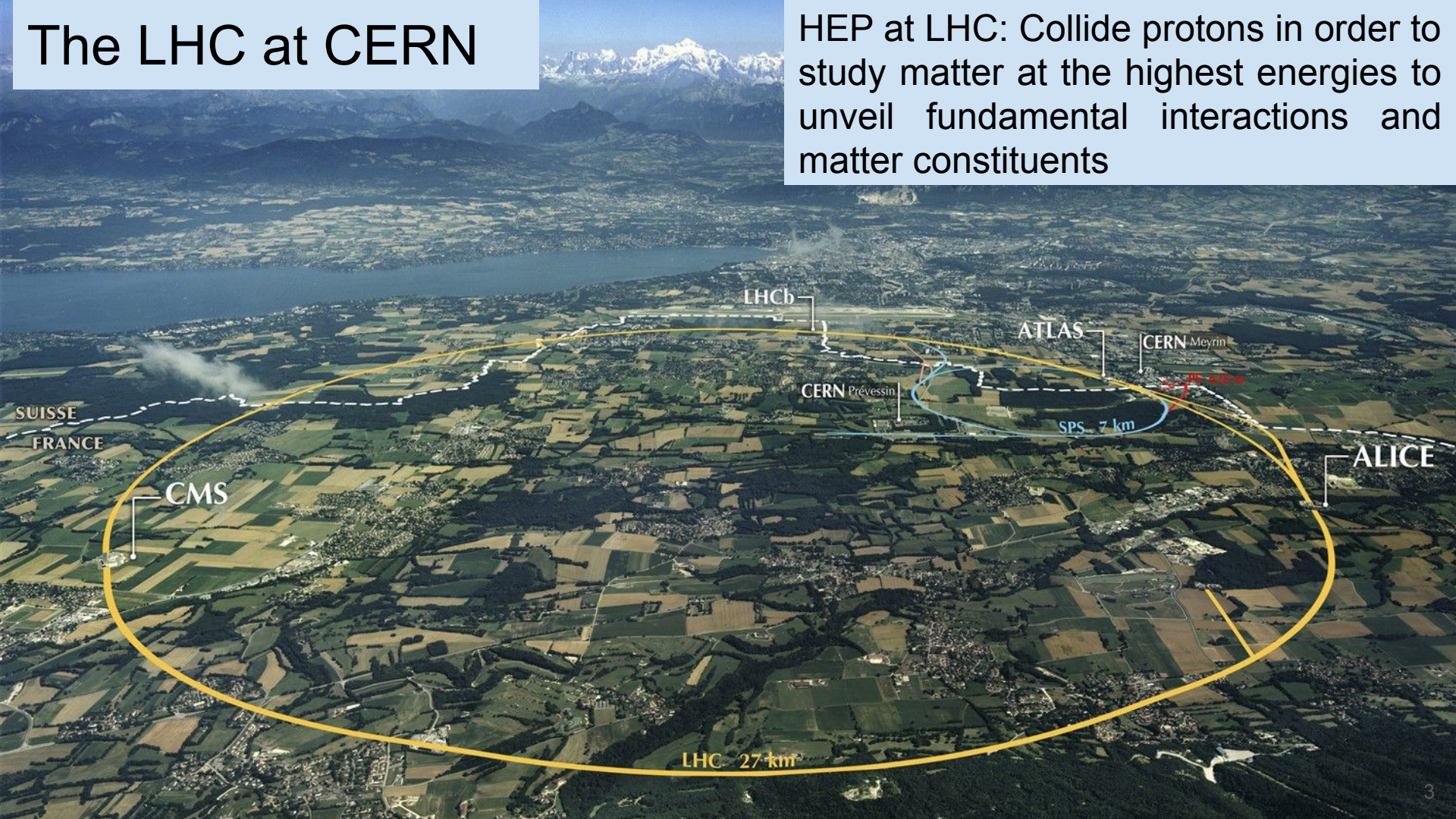
Antonio Pérez-Calero Yzquierdo
PIC Seminar, 20 October 2023

Outline of the talk

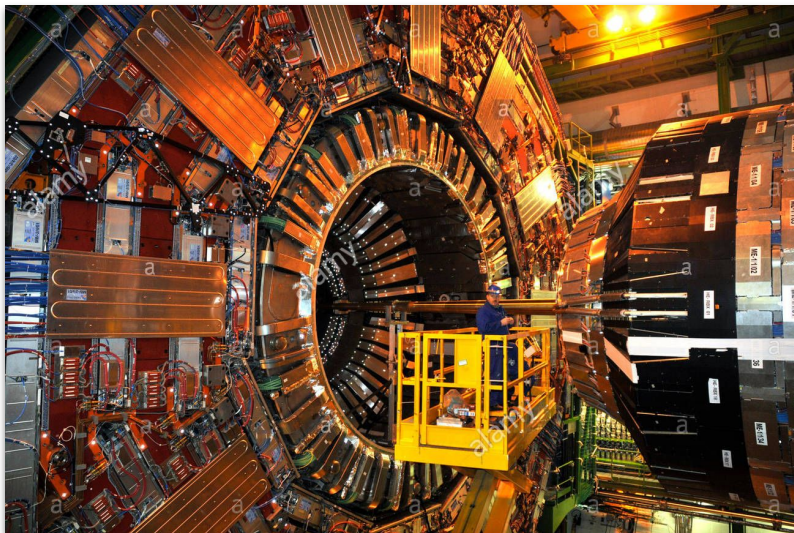
- The CMS experiment at the LHC at CERN
- The Submission Infrastructure in CMS Offline & Computing
- Authentication and tokens
- Integration of HPCs
- Scalability of the SI
- Heterogeneous resources
- The Online farms at P5
- CPU efficiency and pilot overloading
- Conclusions and future work

The LHC at CERN

HEP at LHC: Collide protons in order to study matter at the highest energies to unveil fundamental interactions and matter constituents

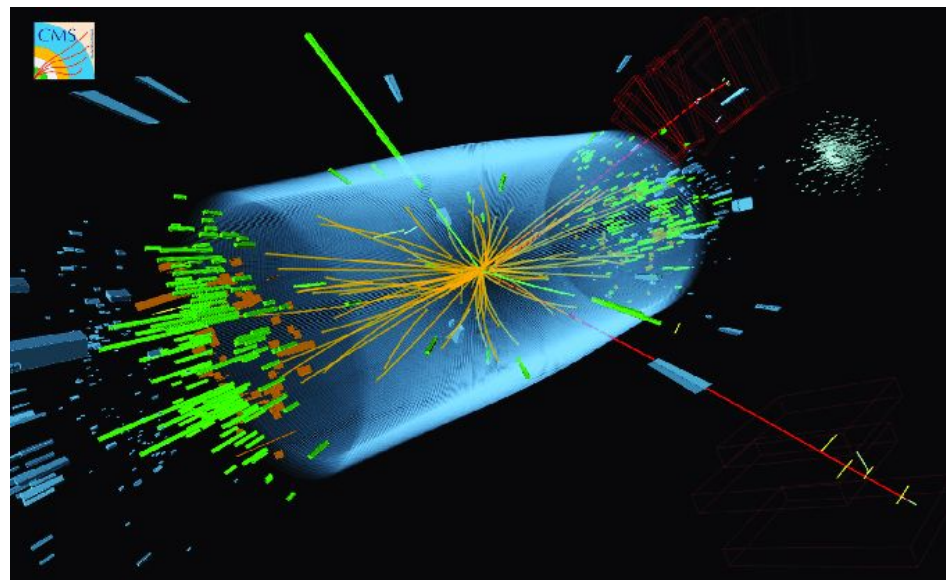


The CMS experiment at CERN



- **Experimental data** is stored, distributed, reconstructed, and analyzed, comparing to **simulated data** (Monte-Carlo)
 - **Hundreds of PBs per year**

- **High Energy Physics general-purpose experiment at the LHC at CERN:**
- **Where the Higgs and other new fundamental particles are being discovered and studied!**



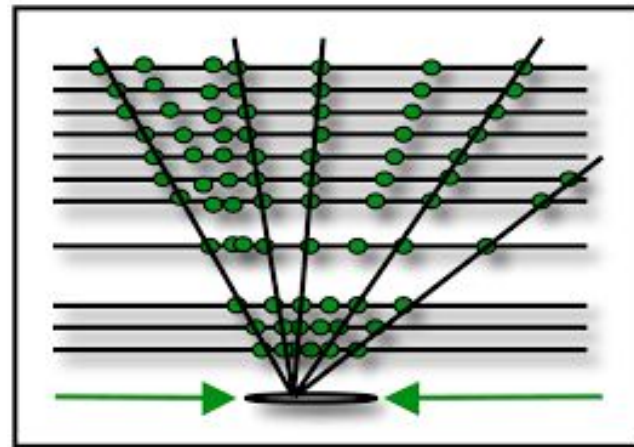
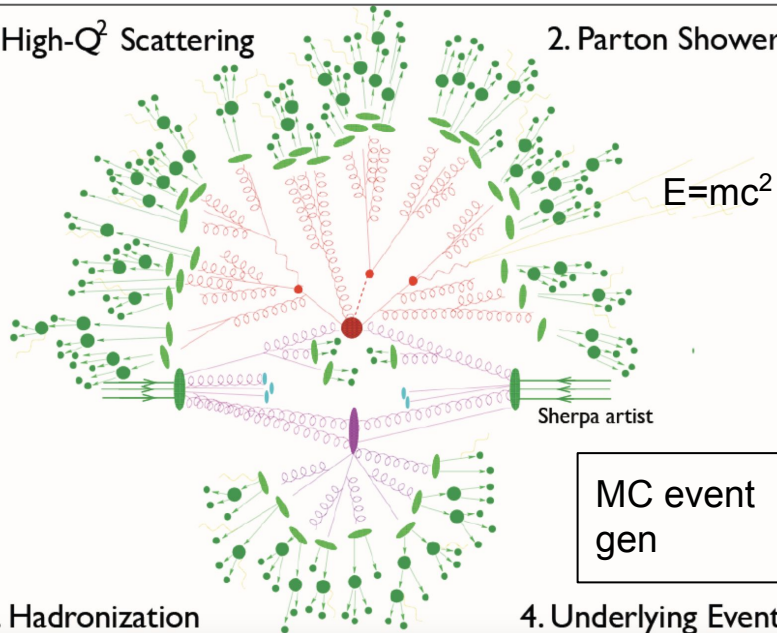
Data simulation, reconstruction and analysis in CMS

CPU time is split between several tasks :

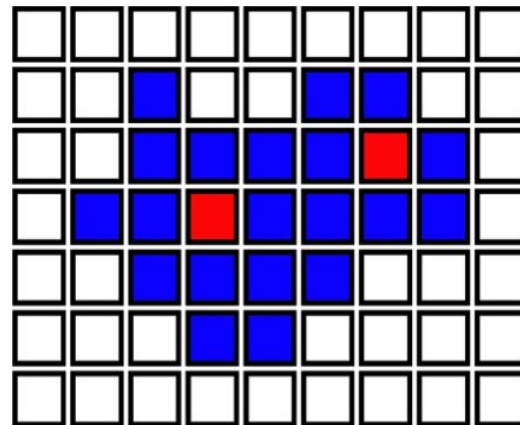
- Event generation
- Simulation
- CMS event reconstruction
- Reduced data formats and analysis

1. High- Q^2 Scattering

2. Parton Shower



Tracking



Calorimeter clustering

The computing landscape - the WLCG

- Computational resources traditionally obtained from the **Worldwide LHC Computing Grid (WLCG)**
 - Global collaboration of around **170 computing centers in 42 countries** -> T1_ES_PIC
 - Access based on dedicated resources (**pledges**)
 - **Nearly 1.4M CPU cores and 1.5 EB in data storage**

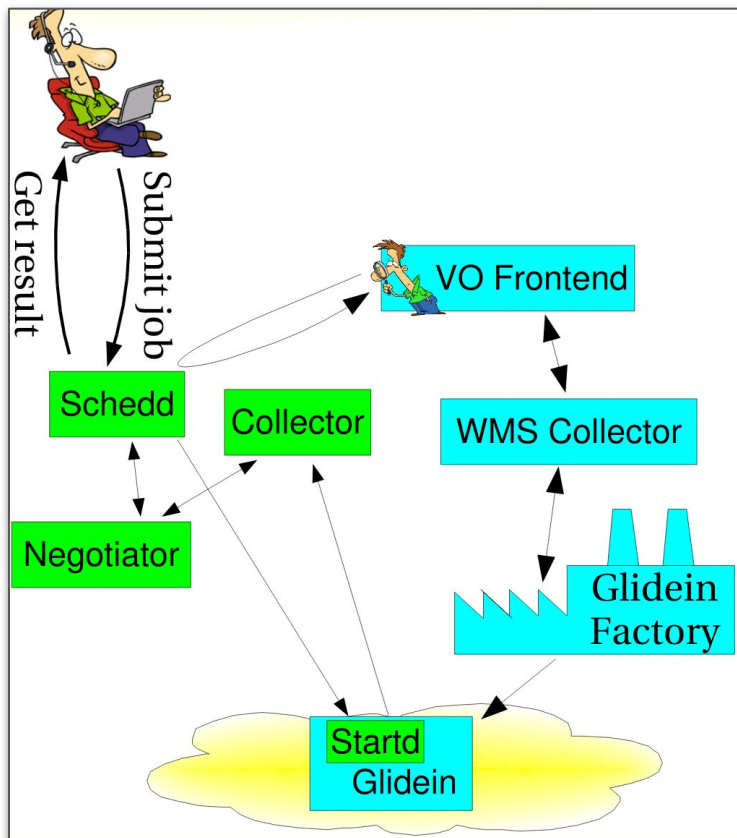


The CMS Submission Infrastructure

The CMS Submission Infrastructure Group

- Part of CMS Offline and Computing in **charge** of:
 - Managing the **main infrastructure** where all reconstruction, simulation, and analysis of physics data takes place
 - Organizing **HTCondor** and **GlideinWMS operations** to sustain this infrastructure
 - Communicate CMS **priorities to the development teams** of **GlideinWMS** and **Condor**
- In practice:
 - We operate a set of federated pool of resources **distributed over 70 CMS WLCG sites, plus non-Grid resources**
 - Join them into a Global Pool of resources managed by HTCondor
- The SI team is deployed at CERN and FNAL, UCSD, in total about 6 members
- I have been a **L2 convener** for this team since January 2016 to date

Resources in CMS dynamic HTCondor pools



Two matchmaking stages in Submission Infrastructure:
resource allocation (GlideinWMS) and job to slot
matchmaking (HTCondor)

1. GlideinWMS and pilot jobs

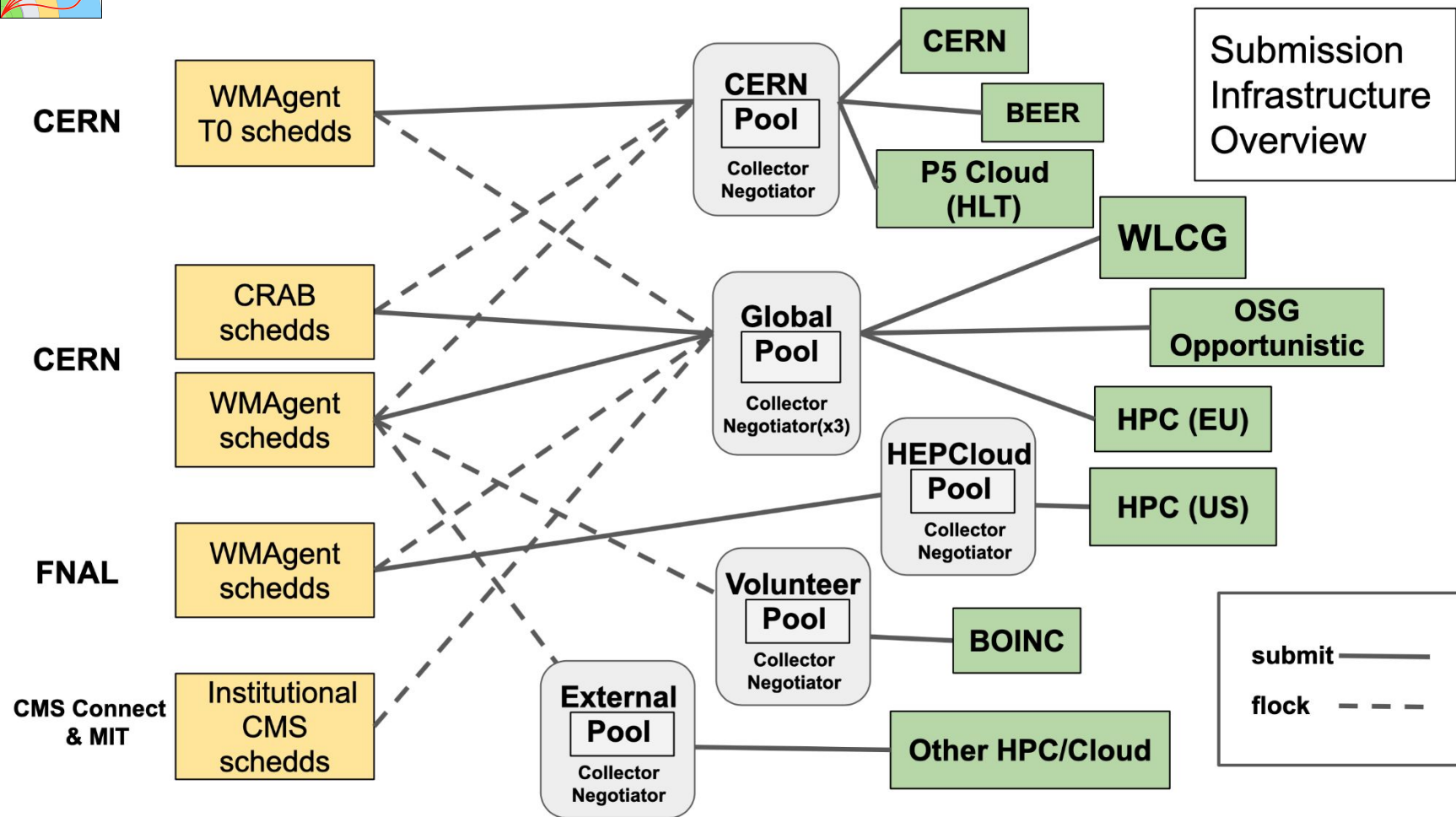
- Submit resource requests on sites CEs in order to join them into the Global HTCondor Pool

2. HTCondor matchmaking of payload jobs

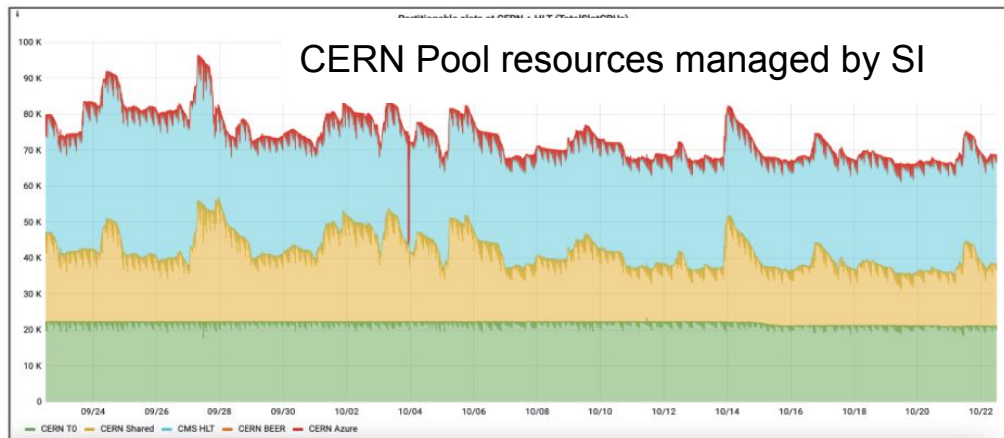
- Slot joins the Global Pool, then resources are negotiated and assigned to payload jobs

Late binding

The CMS SI Overview

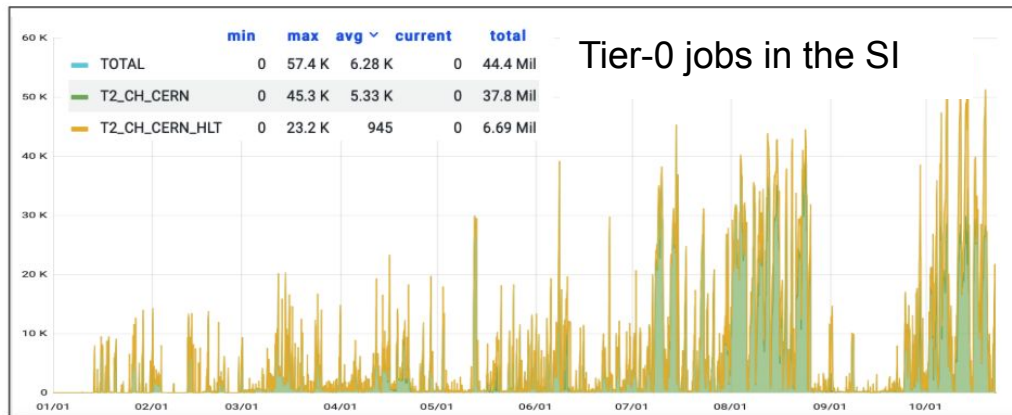


The LHC Run 3 and Tier-0 operations in the SI



As a critical part of its mission in support of CMS, the Submission Infrastructure aggregates and manages all CPU resources available for CMS at CERN:

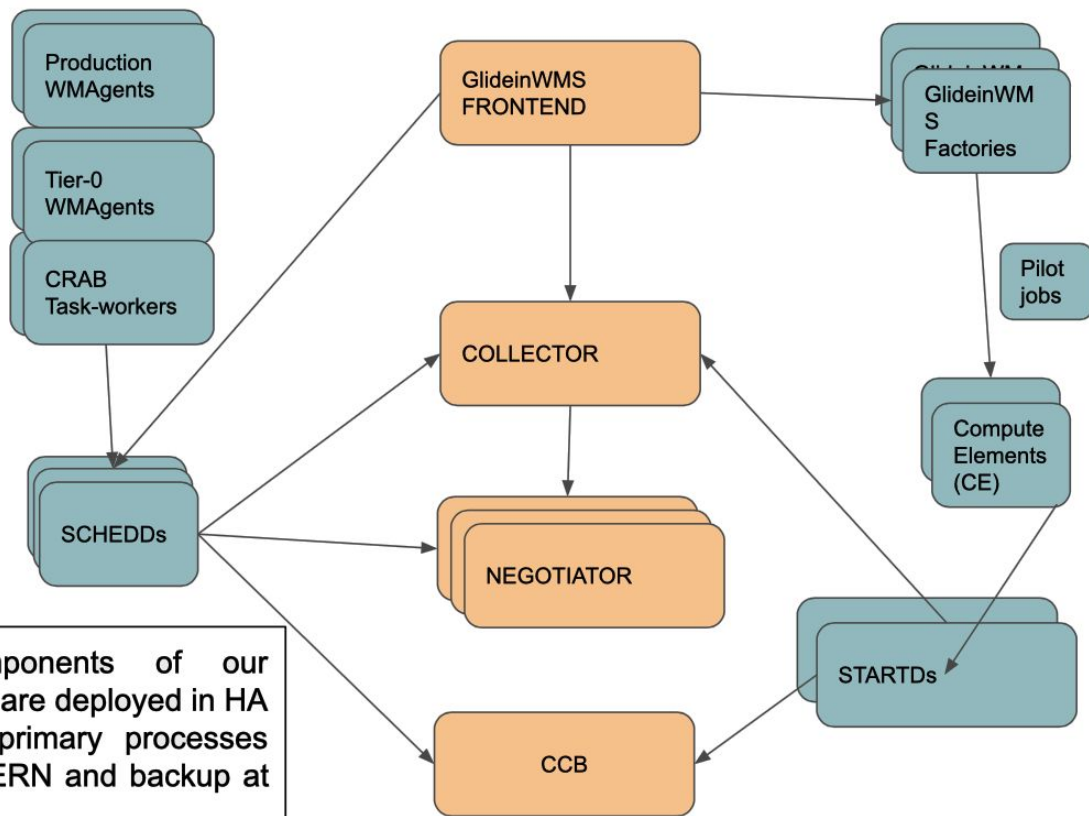
- CERN “Tier0” and “Shared” clusters
- The former Run2 HLT CPUs (“P5 permanent cloud”)
- Opportunistic (BEER) and cloud (Azure) CPUs provided via CERN



Since the start of the LHC Run3, the Submission Infrastructure has been providing resources for Tier-0 tasks (data repacking and prompt reconstruction)

Given the mission-critical role of Tier-0 tasks for the CMS collaboration, several scale tests were performed in the first months of 2022 to ensure a smooth restart of data taking operations. Tier-0 test workloads were injected in bursts to ensure CPU allocation in sufficient quantity and with required responsiveness

CMS Global Pool and stability: **High-Availability deployment** and **redundancy and horizontal scaling of its components**



Critical components of our infrastructure are deployed in HA mode, with primary processes running at CERN and backup at FNAL

Pilot factories are deployed at CERN, FNAL and UCSD, overlapping with OSG's Factory Ops team

Schedulers are also deployed at CERN and FNAL, in numbers sufficient to manage CMS workloads for each major task category: centralized production (MC production and data processing campaigns), analysis tasks and Tier-0 jobs (including prompt data reconstruction at CERN).

Infrastructure management

The SI is built upon a number of machines at CERN and FNAL. This is a summary of the infrastructure and how it is managed:

- **Nodes:** 24 nodes at CERN, 15 at FNAL
 - Schedds are mostly virtual, CM, CCB, FE and factories are physical
 - HTCondor 10.0.1, currently upgrading to 10.0.7
- **Puppet** is the configuration management system of choice at CERN and its used to manage all machines. Hostgroups:
 - `vocmssi/glideinwms/gwmsfrontend` -> GlideinWMS Frontend machines
 - `vocmssi/schedd_htcondor/` -> Schedds, shared with the respective T0, production and CRAB teams
 - `vocmssi/si_htcondor/ccb` -> HTCondor conection brokers
 - `vocmssi/si_htcondor/central_manager` -> HTCondor Central Manager
 - `vocmssi/si_htcondor/condormon` -> Monitoring node, mainly for Grafana
- **Secrets management:** stored in the Teigi secret storage service
 - passwords (aka signing keys) used by Condor and GlideinWMS to generate and verify tokens
 - tokens that are used by the condor daemons to authenticate themselves.
- **Monitoring and alarms:** EL, grafana, etc

Monitoring the infrastructure

A **huge effort** dedicated along the years to improvement of the CMS SI:

- “Back in the day”, CMS only had “job dashboards”, lacking all information concerning the HTCondor pools, CM status, GlideinWMS FE and factories, etc
- A critical aspect of the transition to multicore pilots i(~2016) in order to ensure success in CPU efficiency of our multiple payload per pilot approach
- An essential tool to properly understand and solve our scalability limitations
- Provide a detailed view of the available resource landscape to CMS collaboration
 - Actual amount of CPUs per site, sites under pledge, sites over pledge
 - Specialized resources such as GPUs
 - Enabling IPv6 evolution
 - Resource description (micro-architectures, AMD vs Intel, non x86) and location
 - Host monitoring
 - Infrastructure alerts



Monitoring examples

SI ⓘ

CMS Submission Infrastructure: ALERTS
Production

CMS Submission Infrastructure: Collector monitor
Production

CMS Submission Infrastructure: Factories monitor
Production

CMS Submission Infrastructure: FE monitor
Production

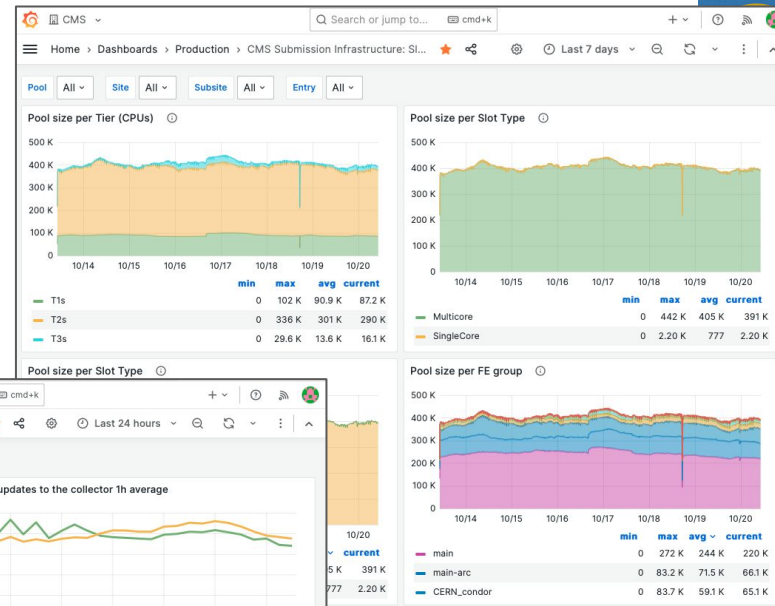
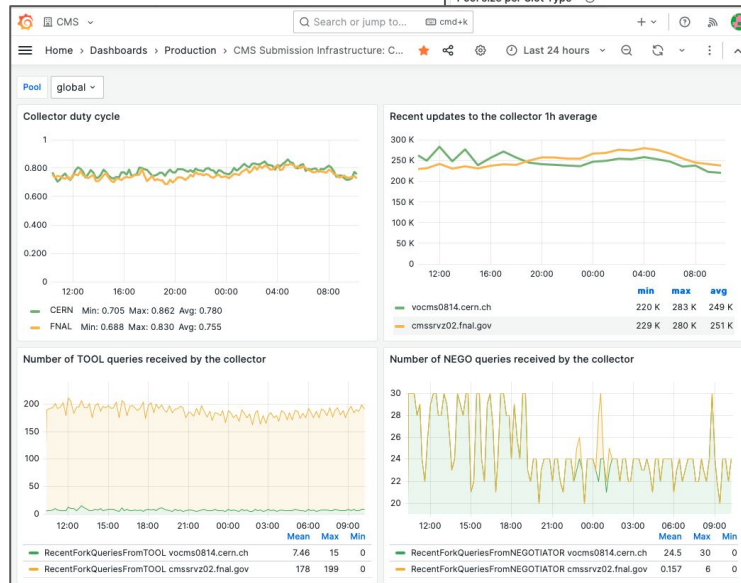
CMS Submission Infrastructure: GPUs monitor
Production

CMS Submission Infrastructure: Hosts monitor
Production

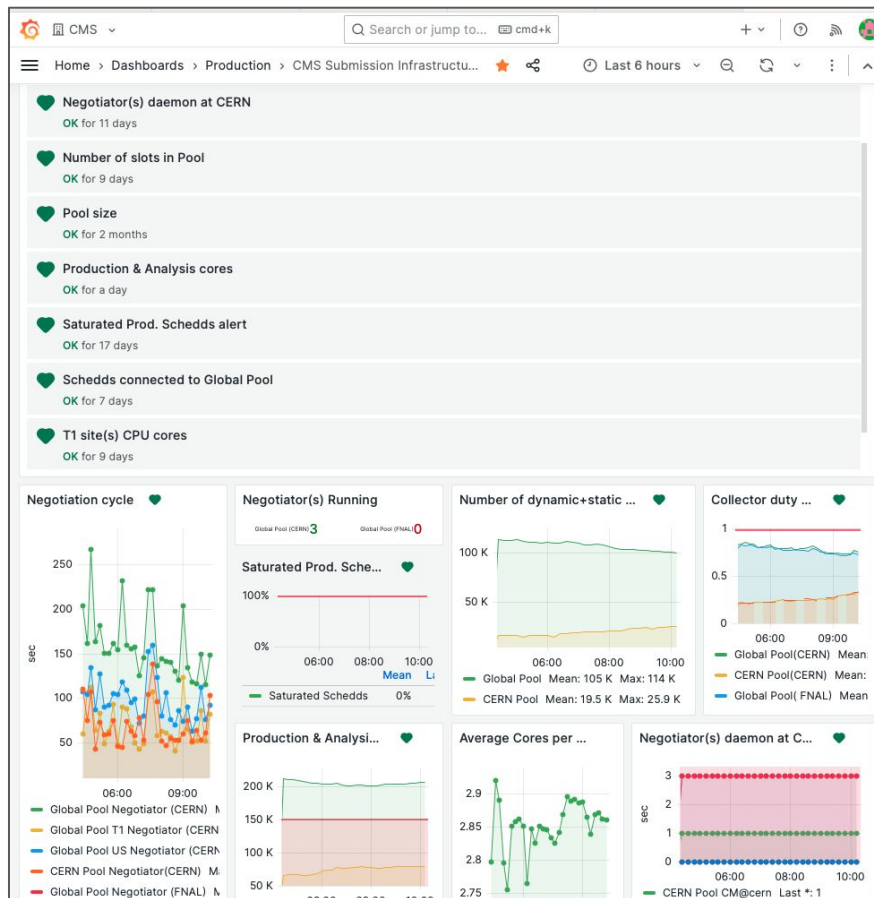
CMS Submission Infrastructure: Negotiator monitor
Production

CMS Submission Infrastructure: Payloads monitor
Production

CMS Submission Infrastructure: Schedd monitor
Production



Alarms examples



The CMS SI “Fire drills” (Spring 2022)

Fire drill exercises:

- **Before the start of the LHC Run3 data taking operations**, the SI team performed a series of **exercises** to validate the effectiveness of safety mechanisms embedded in its design
- In these exercises (“**fire drills**”) critical **services** associated with both the Global and CERN pool infrastructures were intentionally **disabled**, in order to force secondary services reaction

Conclusions:

- **The SI system is stable and fault tolerant** in the context of any one of the central components (GlideinWMS Front-End and HTcondor collector, negotiator or CCB) accidentally becoming unavailable, with **no impact on the overall performance** of the infrastructure (as expected!).
 - In some cases, **second-order effects** were detected (mainly in relation to the service **monitoring** the SI), followed by corrective actions
- Still, a number of **secondary sources of potential instability** have been identified in external dependencies and should also eventually be tested.
 - For example, an outage of the GIT repositories storing the configuration information for each of our services cause additional delays in the deployment of new hosts as recovery measure to counter the loss of any main pool element
 - Several SI services as hosted by VMs (e.g. schedds) whose configuration is located in separated CEPH disk volumes, presumably affected in case of outages

Authentication and tokens

Why tokens?

M.4	Mar 2023	HTCondor installations at EGI sites have been upgraded to supported versions > 9.0.x. Prerequisites: • DIRAC versions supporting job submission tokens have been deployed for the concerned VOs (LHCb, Belle-II, EGI catch-all, ...) • HTCondor CE supports (adjusted) EGI Check-in tokens • IAM or equivalent in production for ALICE, LHCb, Belle-II, ...	M.1	HTCondor Dev Team, WLCG ops, EGI ops, sites
M.5	Mar 2023	End of HTCondor support for GSI Auth (link).		

Motivations:

- General **WLCG [campaign](#)** to use the **de facto standard** in place of **Grid specific ones (GSI)**

Advantages:

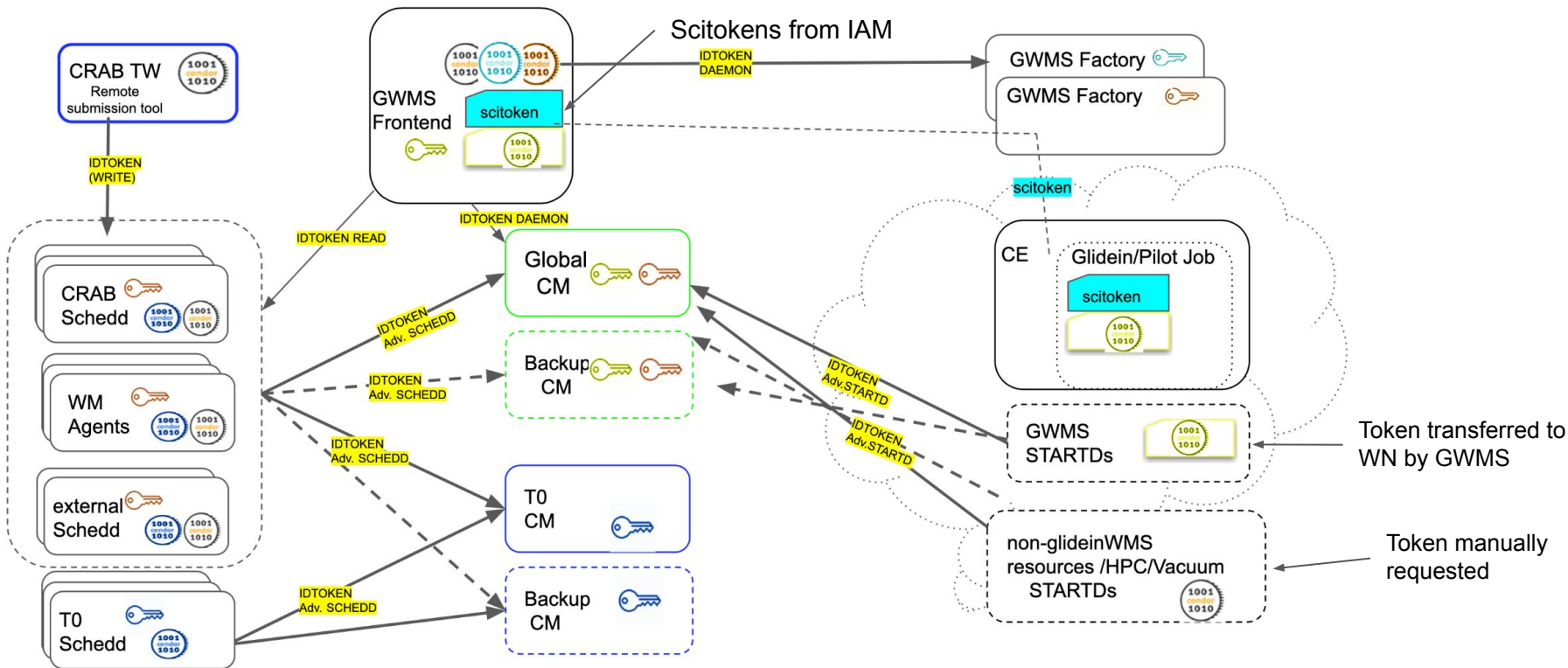
- Authorization based on token **capability** (vs identity) allows for finer handling of communications between entities
- Easier to follow **principle of least privilege** (**limited impact** in case of credential leak, for example)

Technologies

- **IDTOKENS** for HTCondor Software Suite components, **native** condor authentication mechanism
 - JWT tokens signed with a **simmetric key**
 - Tokens generated with condor **command line interface** (even remotely)
- **SCITOKENS** for Grid sites (HTCondor-CEs + ARC-CEs)
 - Uses WLCG IAM go generate tokens: a client is registered and can fetch tokens on the GWMS Frontend
 - JWT tokens signed using **private/public schema**
 - WLCG CEs verify authenticity of tokens and authorize pilot job submission

Overall authentication in SI

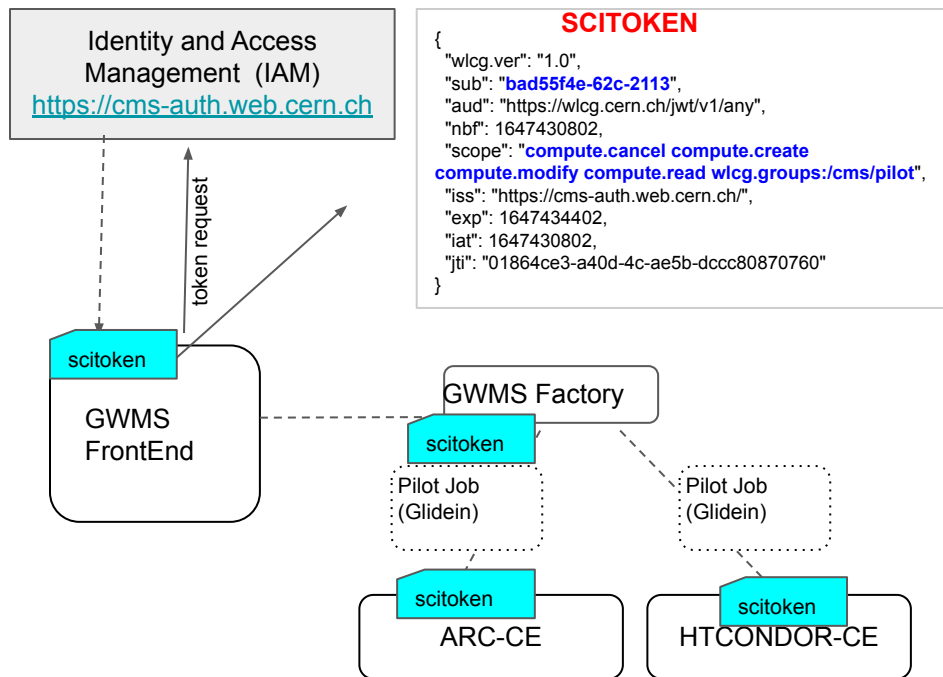
- The whole CMS Global Pool is using IDTOKENS for authentication
 - **HTCSS** version 10 with **no GSI fallback** installed early April
- Multiple keys and tokens to **minimize impacts** in case of security incidents



Submitting CMS jobs to the Grid

GlideinWMS Factories submit pilots jobs using HTCondor to WLCG sites, which employ **two CE technologies**:

- **HTCondor-CE:** Completed 100% (both OSG and EGI sites (last site January 2023))
- **ARC-CE:**
 - Submission through LDAP interface and GSI proxies
 - Deprecated in HTCondor 10 series
 - New REST interface available
 - Support for **tokens**
 - Plain X509 proxies can be used (no GSI)



Integration of HPCs

HPCs for CMS Computing

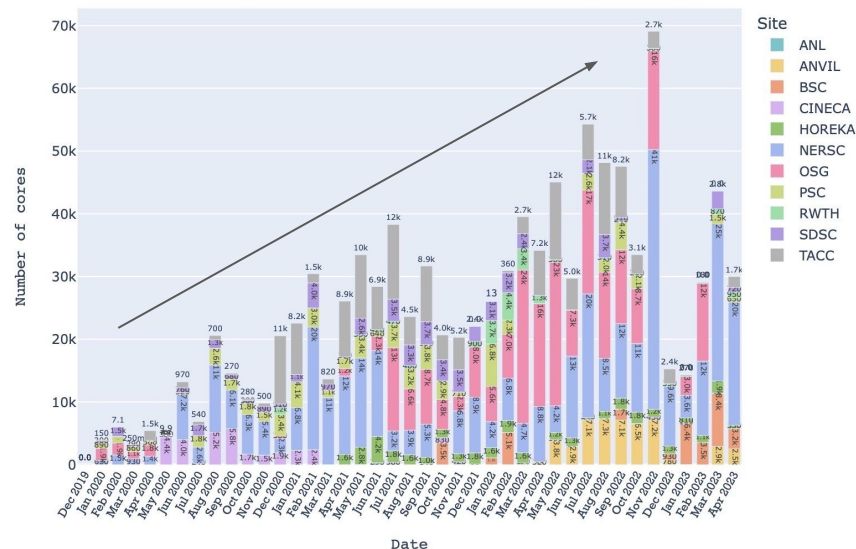
HPCs can help supporting the computing needs of CMS:

- Substantial national and supranational investments made and planned
- HPCs are part of the scientific computing infrastructure, and there to stay
- Several HPC integration efforts in CMS

It appears that the **compute capacity used by CMS at HPC facilities** could be **maintained** at the current level, **if not higher**, in the future years.

CMS Public

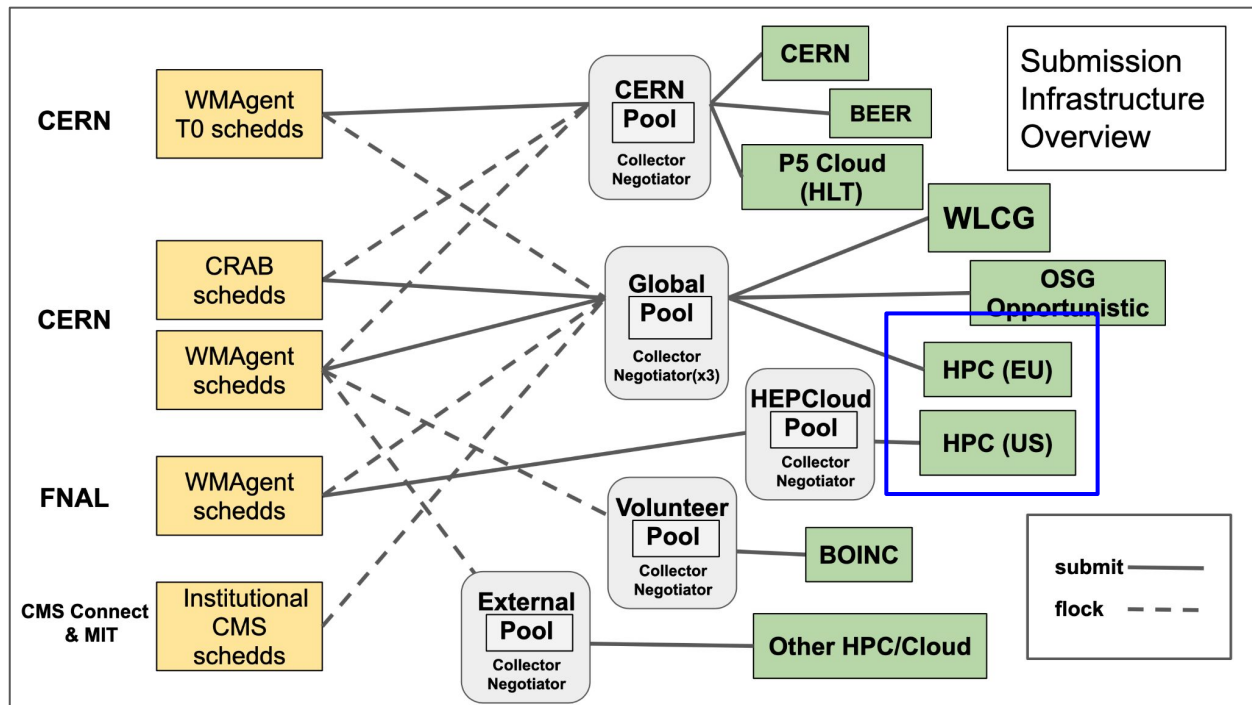
Number of Running CPU Cores on HPCs - Monthly Average



Integration of HPCs: two main mechanisms

Two main models of HPC integration to CMS computing:

- **HEPCloud** model for US resources, managed from FNAL: NERSC, ANL, SDSC, PSC, TACC.
- **Transparent WLCG site-extension**, employed for the EU HPC facilities: CINECA (and soon also Vega) via CNAF, BSC via PIC, Jülich and HOREKA via KIT, etc.



Challenges in HPC integration

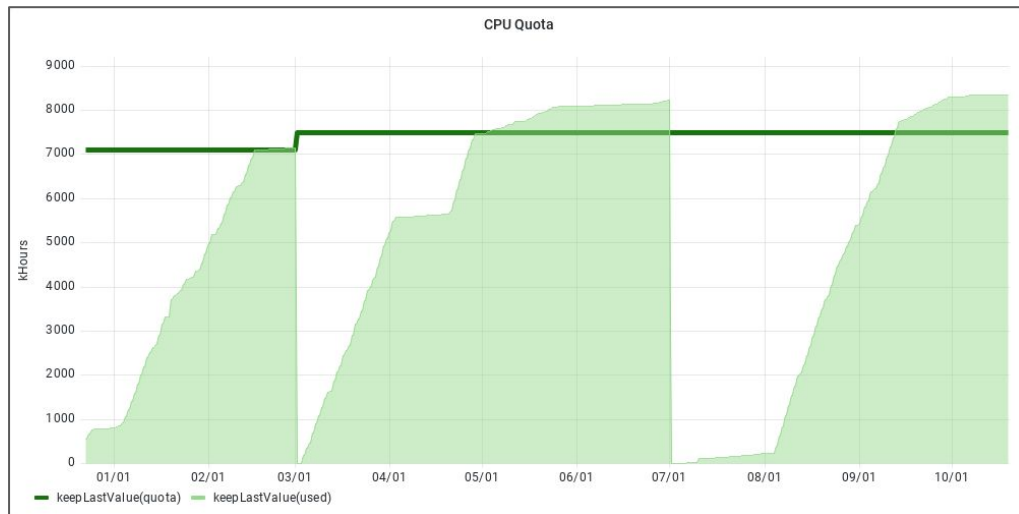
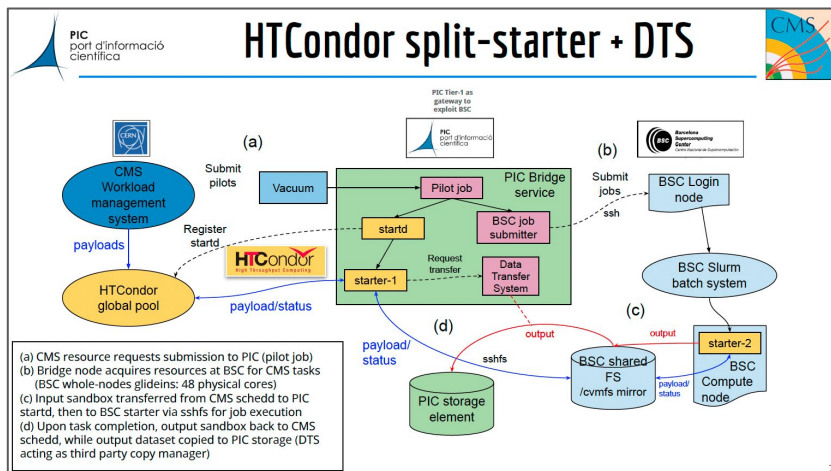
- **Each HPC is different**
 - Diversity in technical and policy constraints (networking, security, edge services, etc)
- **Negotiate** deployment of services **case by case**
 - Singularity, CVMFS
 - Data cache services (for application software, conditions data, event data)
 - Job gateway (CE)
 - http/gridFTP/xrootd doors for input/output data files
- **Configuration** in our WM/DM systems
 - Processing and Storage site names, site config files, trivial file catalogue rules, glideinWMS factory entries, etc
- **Scalability**: discussed later in the talk

The BSC case: success!

The BSC case is one of the most complicated HPC integration tasks faced by CMS, due to the lack of connectivity between the nodes and CMS central services and SEs

BSC integration start model (CHEP'23)

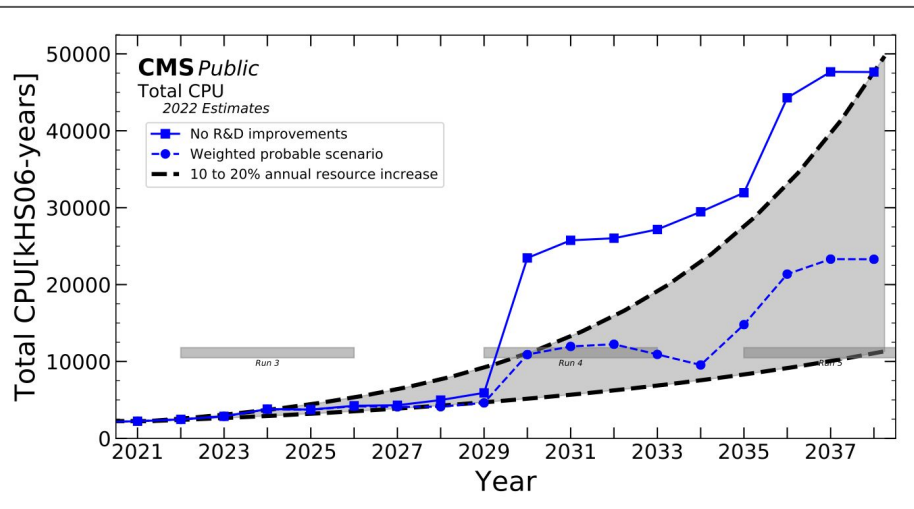
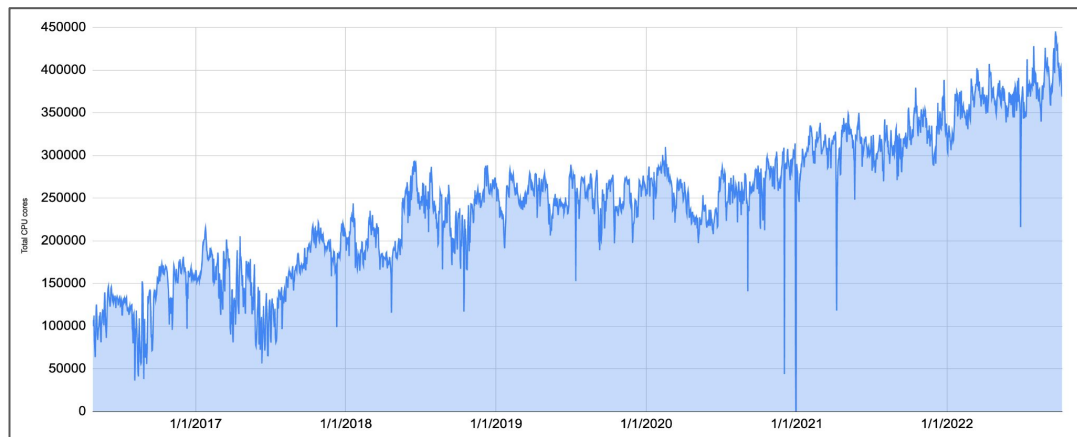
CMS utilization of BSC assigned resources in 2023



Scalability of the SI

The scalability challenge (I)

- Recent past: continuous growth in the number of CPU cores allocated to CMS over the past ~6 years (daily averages)



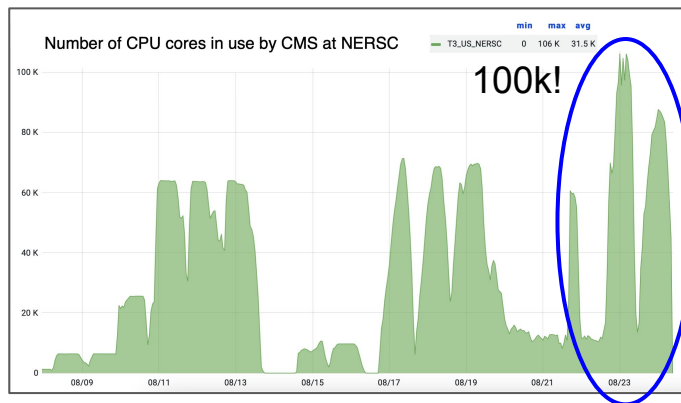
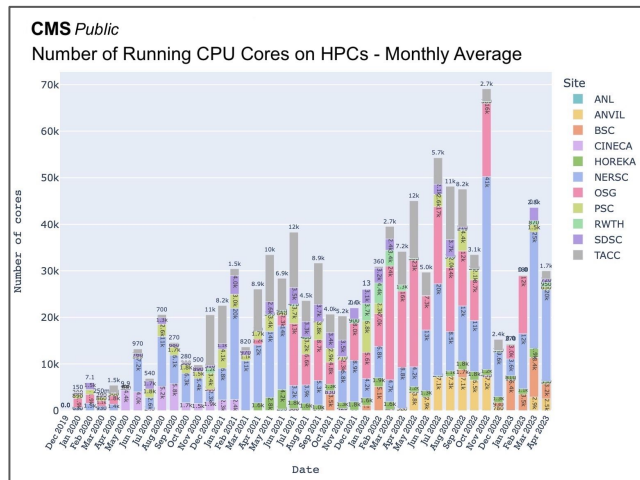
- Future: need of resources expected to grow, especially in the next High Luminosity phase of the LHC program (HL-LHC, starting 2029)

The scalability challenge (II): HPC component

Scalability challenges associated to HPCs resources exploitation:

- **Sufficient WM and SI capacity** to manage and supply workloads to fill increasingly large pools of resources
- **HPC vs WLCG provisioning:** WM and SI capacity should be dimensioned to profit from HPC **resource peaks**, in addition to WLCG resource baseline

Stability and flexibility of the WM infrastructure and services essential to ensure effective use of CMS compute capacity enlarged with HPCs



The scalability aspect of our work in the CMS SI

- **The CMS SI team makes sure we operate our infrastructure away from any scalability limiting factor**
 - **Critical aspect** for a system that is designed to perform in a **dynamic environment**, adapting itself to growing **resource demands** by CMS, **resource availability** in the WLCG and the **mix of workloads** it has to manage
- **Proactively find those limits**, in every direction, and evolve the infrastructure to push them further away:
 - **Total computing power** our HTCondor pools can harness and use efficiently
 - **Collector capacity** to process the stream of **slot updates** and keep resource status fresh
 - **Negotiator matchmaking** cycle time under control
 - Total **number of workflows** we can manage and jobs we can run simultaneously with our **pool of scheduler nodes**

Pushing the limits of the CMS Global Pool

- Since the start of the Run 2, we have been continuously **detecting and solving bottlenecks to our Global Pool**, with the support of the **HTCondor** and **GlideinWMS** developers teams over the years
- **Multiple customized settings** compared to a “standard HTCondor configuration”:
 - Using a **CCB** running on a separate host to the CM, with enlarged pool of connection sockets
 - Use of multiple **negotiator** daemons running in multithreaded mode
 - 32-bit binaries for the **shadow processes** running in the schedds (presently disabled)
 - Hierarchy of **secondary collectors** connected to the main top collector process
 - Optimized **slot update conditions** (filter on update triggers, use UDP instead of TCP, enlarged UDP buffer)
 - **Classify queries** reaching the collector from the negotiator as high-prio, as opposed to those from the GlideinWMS FE and CMS WM and monitoring services
 - **Redirect non-high prio queries** to the backup secondary collector (HA infrastructure at FNAL)
- Most of these actions directed at **avoiding the saturation of the top collector**
- Essential to build an **exhaustive and reliable monitoring infrastructure**
- **Successive rounds of our CMS SI scalability tests reported at CHEP:**
 - Reaching new peaks for the future of the CMS HTCondor Global Pool, vCHEP 2021, <https://doi.org/10.1051/epjconf/202125102055>

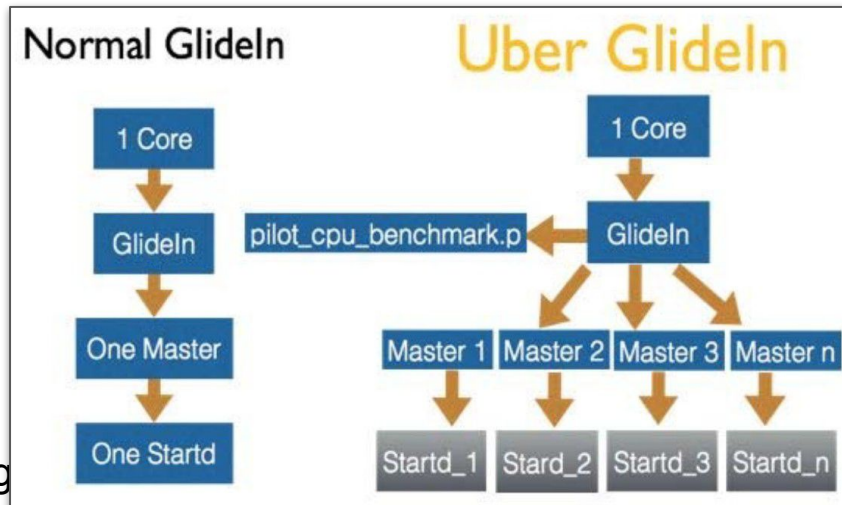
The SI 2023 scale tests

Following updates since our latest tests (2021):

- Evolution in HTCondor software
 - Tested version 10.0.1
 - New feature for optimizing slot updates
- A **new physical host** for the central manager
 - AMD EPYC 7302 at 3 GHz
- **Token**-based

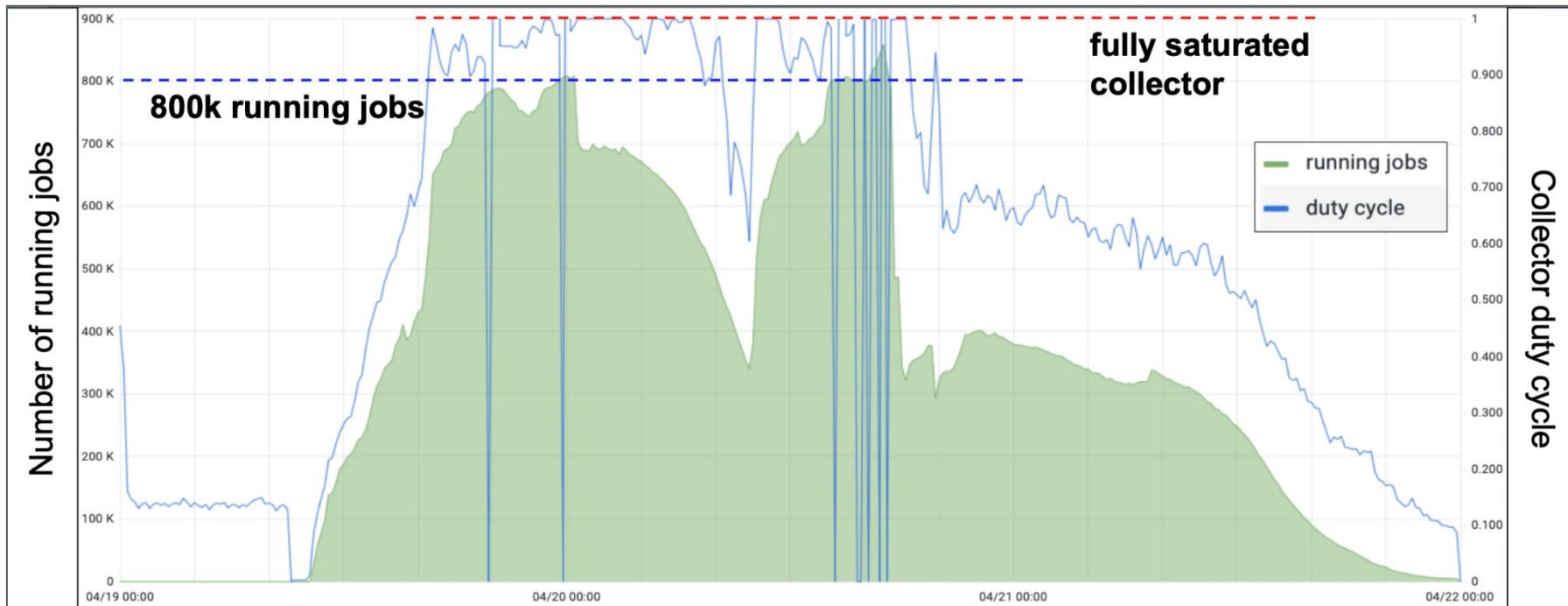
As in previous tests

- Über-glideins used to g
- Secondary collector service in the pool as a source of monitoring data, to reduce stress from non-essential queries on the main collector



The SI 2023 scale tests: results

- After some initial bottlenecks detected in a) the pool of **schedds capacity**, and b) TCP connections in the **CCB**, we managed to push the scalability of our Global Pool to about **800k simultaneously running jobs**
 - Factor of 5 - 8 away from usual number of running jobs observed in our pool (100k to 150k)



Heterogeneous resources

Interest on heterogeneous resources

- **The availability of compute power in non CPU, non x86 processor types is abundant and increasing**
 - For example, looking at the examples from the top500.org list, it's clear that:
 - A significant fraction of the processing power in HPCs is provided by accelerators.
 - Many top positions include non-x86 CPU architectures (e.g. IBM Power systems:Summit, Sierra, Marconi-100)
 - Not only at HPC facilities, but also among traditional WLCG computing sites as well
- **HEP experiments must be prepared to run substantial part of their processing tasks at HPCs**
- CMS internal study (**ECoM2x, 2020**) of the computing model and resource needs looking into the LHC Run 3, but specially at the HL-LHC phase, includes in its recommendations:
 - *"CMS should continue to **aggressively expand the resources** accessible to it for production processing at facilities beyond those dedicated to the LHC"*
 - *"**CMS should strive towards using HPC resources effectively**"*
 - *"Use of on board **accelerators** as much as possible during reconstruction"*
 - *"Work towards enabling CMS software on CPUs, GPUs, FPGAs and TPUs as primary targets for its software stack"*
 - *"**Support non x86_64 CPU architectures**"*
- The CMS Phase-2 Computing model update document (**CMS-NOTE-2022-008**) also advises to **abandon the assumption of uniformity of our compute resources, evolving our WM and SI systems to embrace heterogeneity**

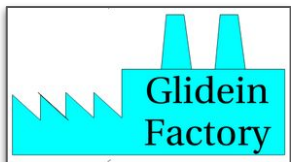
Non-pledged heterogeneous resources

In relation to the use of GPUs in CMS offline computing, we need to consider that all GPUs available and in use by CMS today are still opportunistic, not pledged:

- **Pledged CPUs: agreed upon standard job execution slot**
 - 8-core slots, 48h of runtime, minimum of 2 GB/core, etc
- **No equivalent “standard job slot” for GPUs.** This affects:
 - How to approach sites on what GPU resources CMS can use: how do we even know there are GPUs available for CMS? What CEs, queues and other parameters to use?
 - What is the correct GPU type for CMS tasks?
 - How to configure the execution of tasks on these resources: no predefined rules on e.g. slot size, max execution time, memory, etc
- **Deeply affects how we use the resources:** no generic slot, requires a more careful and detailed job and slot description in order to select the correct slot and avoid wastage

GPU resource description in GWMS Factories

- GPU resource description for first matchmaking in pilot factory entries
 - Limited information about the available GPUs:
 - CEs that allow access to GPUs, queue names, etc
 - Slot attributes **agreed upon** with the site admins, (e.g. GPU slot lifetime 24h? 48h?)
 - **Statically configured, with most GPU specs only available at pilot runtime**
 - Typically, GPU slots are configured along with standard pilot features (8 CPU cores, 2 GB/core RAM)



```
<entry name="CMSHTPC_T2_US_Wisconsin_cmsgrid01_gpu" auth_method="grid_proxy"
gatekeeper="cmsgrid01.hep.wisc.edu cmsgrid01.hep.wisc.edu:9619" gridtype="condor" ...>
  <config>
    (...)
    <submit_attrs>
      <submit_attr name="+maxMemory" value="20000"/>
      <submit_attr name="+xcount" value="8"/>
      <submit_attr name="Request_GPUs" value="1"/>
    </submit_attrs>
  </submit>
</config>
<attrs>
  <attr name="GLIDEIN_CMSSite" ... type="string" value="T2_US_Wisconsin"/>
  <attr name="GLIDEIN_CPUS" ... type="string" value="8"/>
  <attr name="GLIDEIN_MaxMemMBs" ... type="int" value="20240"/>
  <attr name="GLIDEIN_Max_Walltime" ... type="int" value="216000"/>
  <attr name="GLIDEIN_Resource_Slots" ... type="string" value="GPUs,1,type=main"/>
  (...)
</attrs>
</entry>
```

Resource description in the Global Pool

- **Once a pilot starts** execution on remote resources, **GPU properties** are updated to the slot classad:
 - **condor_gpu_discovery** retrieves most of the relevant matchmaking attributes
 - **CMS_CUDA_SUPPORTED_RUNTIMES** from CMS script in cvmfs

GPU slot attributes

```
CPUs = 8
TotalSlotMemory = 20000
GPUs = 2
CUDACapability = 8.0
CUDAClockMhz = 1410.0
CUDAComputeUnits = 108
CUDACoresPerCU = 64
CUDADeviceName = "NVIDIA A100-PCIE-40GB"
CUDADriverVersion = 11.3
CUDAECCEEnabled = true
CUDAGlobalMemoryMB = 40536
CUDAMaxSupportedVersion = 11030
CMS_CUDA_SUPPORTED_RUNTIMES = 10.1,10.2,11.0,11.1,...
CMS_NVIDIA_DRIVER_VERSION = 515.48.07
```

Second matchmaking in HTCondor

- For each properly configured GPU slot that joins the Global Pool, the HTCondor *negotiator* will compare **job requirements** with machine attributes (**second matchmaking**)
 - Any dynamically retrieved GPU property can now be used to filter suitable slots in relation to a given workflow

Job requirements

```
RequestGPUs = 1
RequiresGPU = 1
...
Requirements =
CUDACapability >= 3 &&
CUDARuntime = "11.4" &&
GPUMemoryMB = 8000 && ...
...
```

Collector + Negotiator

```
...
Machine.CUDACapability in Job.CUDACapability
Job.CUDARuntime in Machine.CMS_CUDA_SUPPORTED_RUNTIMES
Job.GPUs <= Machine.GPUs
...
```

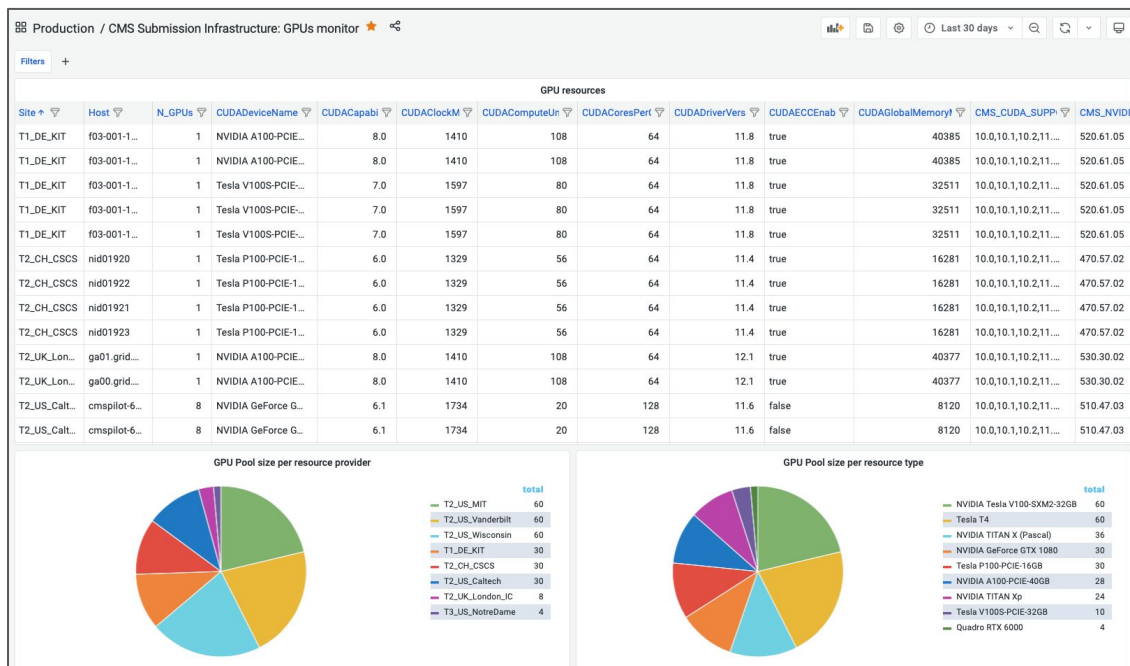
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CUDADriverVersion = 11.3
CUDAECCEEnabled = true
CUDAGlobalMemoryMb = 40536
CUDAMaxSupportedVersion = 11030
CMS_CUDA_SUPPORTED_RUNTIMES =
10.1,10.2,11.0,11.1,...
...
```

- With regards to **how CMS jobs relate to GPUs**, while the situation is evolving, three use cases are considered:
 - Job **must** use GPUs
 - Job **can** use GPUs
 - Job **can only** use CPUs
- **Access and use of GPUs** under control of two main attributes in the job jdl:
 - **RequiresGPU**: Defines whether or not GPU pilots will be requested by the first (GlideinWMS FE) matchmaking
 - **RequestGPUs**: Number of GPU resources to be matched in the second (HTCondor) matchmaking
- The three use cases are covered:
 - `RequiresGPU = 1 && RequestGPUs>0` will trigger GPU pilot, then match the slot
 - `RequiresGPU = 0 && RequestGPUs>0` will not trigger GPU pilots but **CAN** match already available GPU slots.
 - Need to discuss preference for CPU+GPU over CPU slots
 - `RequiresGPU = False && RequestGPUs=0` for purely CPU task

Advertising GPUs already available to CMS

- Multiple CMS WLCG sites already kindly making their GPU resources available for CMS use
- **GPU inventory:** SI regularly scanning the Global Pool for GPU resources, collecting their availability and properties, then published in the [GPU monitoring table](#)
 - Recent (e.g. last 30d) results should inform users about “what is available and where”
 - **PIC recently added to this inventory**



Multi-architecture matchmaking (I)

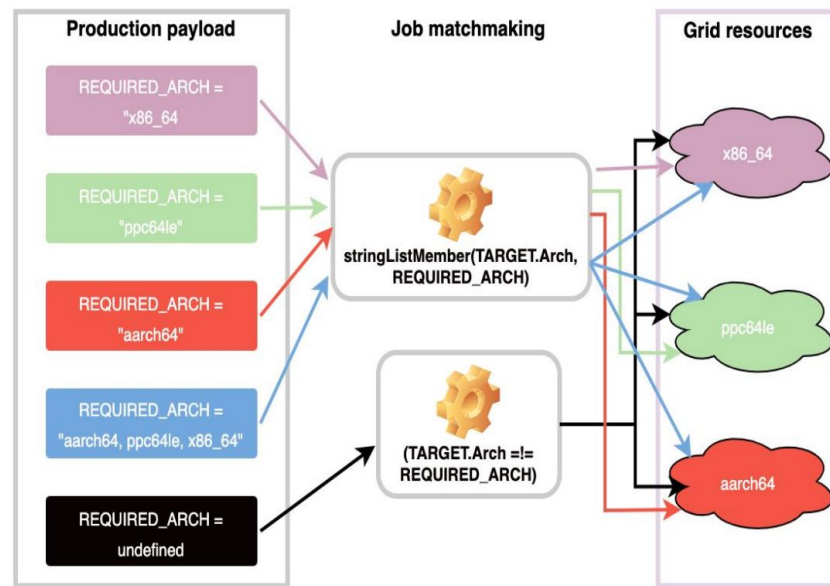
The **CMS SI and WM systems** were originally **built and deployed assuming the x86_64 standard**

- **The WM and SI underwent adaptation** to allow access and use of other architectures
 - **Power9** was the first non-x86 platform available at scale for CMS (at CNAF/CINECA)
 - Initially tested with manually launched pilots (ad-hoc compiled HTCondor startd)
 - no singularity container
 - By mid 2022, HTCondor support for ARM (aarch64) and Power PC (ppc64le) CPU architectures available
 - Startds added to the GlideinWMS pilot factory, can now access these resource types with GlideinWMS pilots
- **Power9 arch is now fully commissioned** and its CMSSW has been physics-validated.
- **Next is ARM:**
 - CERN granted access to a few ARM machines to be accessed as Grid resources, and again at CNAF
 - Also at Glasgow (ATLAS T2)
 - Ongoing **integration** and **physics validation** tests

Multi-architecture matchmaking (II)

Resource architecture has been turned into a configurable **parameter**, to be used in the two CMS SI matchmaking phases:

- Payload jobs describe their resource requirements, which **may** include matching a **specific or multiple architectures**
- Provisioning of the corresponding resources via the HTCondor Global Pool
- HTCondor second matchmaking step



```
Requirements = (stringListMember(TARGET.Arch,"ppc64le,X86_64")) && (TARGET.OpSys == "LINUX") && (TARGET.Disk >= RequestDisk) && (TARGET.Memory >= RequestMemory) && (TARGET.Cpus >= RequestCpus) && (TARGET.HasFileTransfer)
```

Next steps in the use of heterogeneous resources by CMS

Slide shown at the
Fall CMS Offline &
Computing Week, **2**
weeks ago



GPU workflow is not new anymore, what we are talking is to move it to general production level (part of Heterogeneous production, not only GPU)

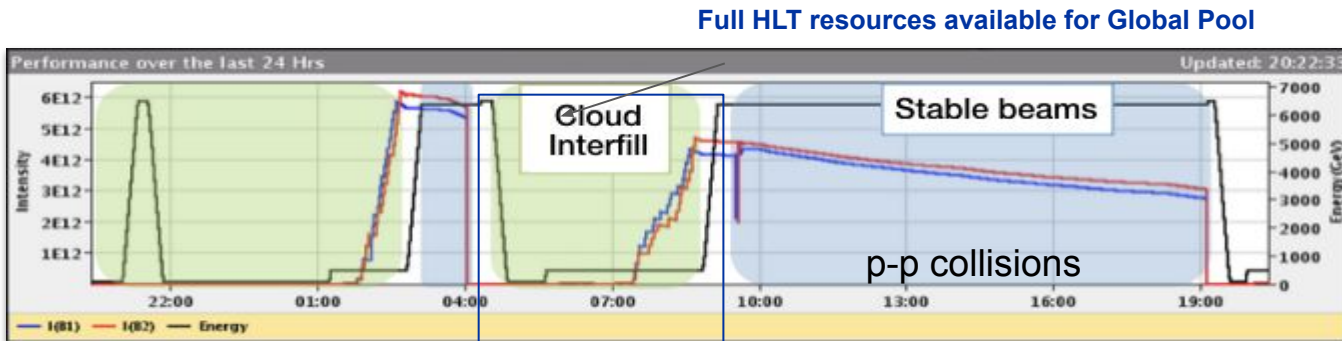
- Software development, more offload to GPU
- Physics validation
- Submission infrastructure + Resources
 - List of site resources, for example, ARM, AMD GPU
 - Use of GPUs in CMS MC production multi-step workflows
 - GPU benchmark, GPU performance, predictable workflow runtimes

The Online farms in P5 for Offline Computing

The HLT farm as part of CMS offline computing

- The **CMS Online Cloud** (deployed on the HLT farm resources) was **commissioned during Run 2** to dedicate **HLT resources for offline data processing** when not needed to support data taking ([ref](#))
- Launch **Openstack VMs** running the HTCondor startds when CPU is available (**Interfill modes**)
 - VMs suspended when HLT needs resources back
 - A **MaxHibernationTime** delay (24h) applied at both CCB and WMAgents schedds in order to ensure job execution can be resumed when the VM is back into the Global Pool

INTERFILL

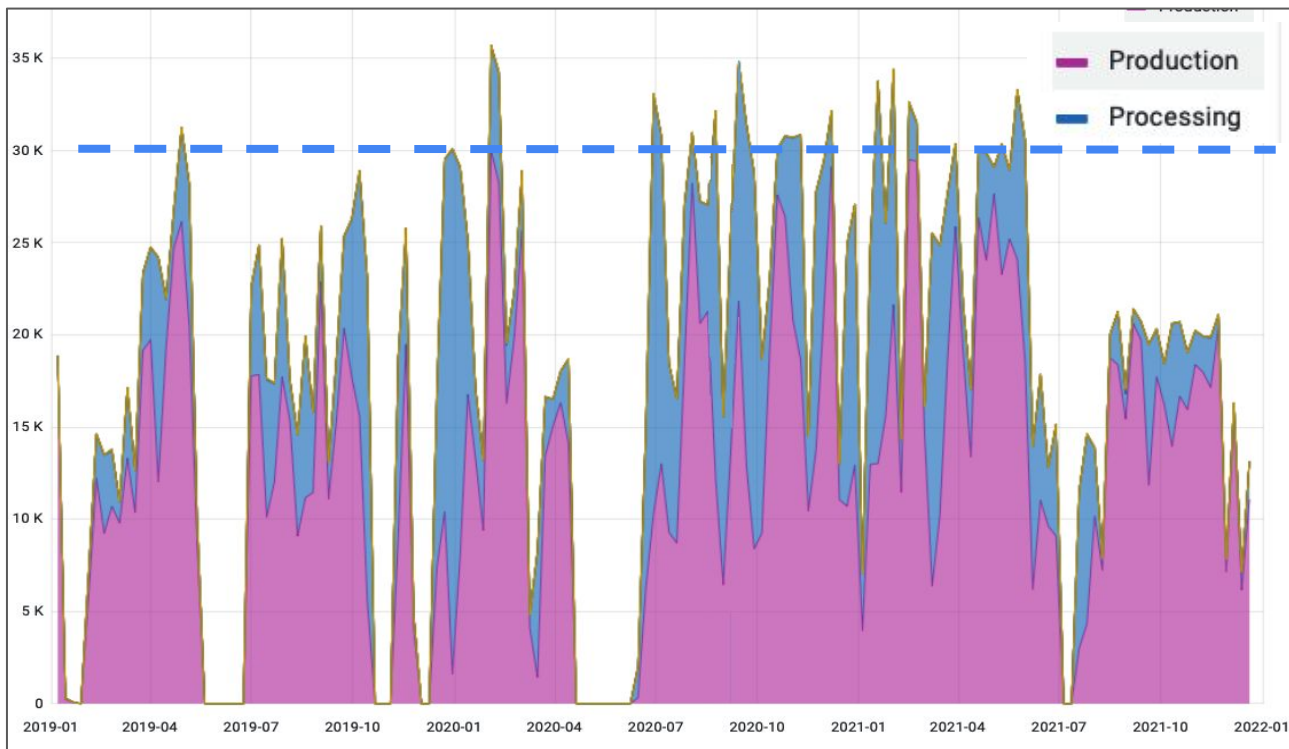


LHC
operation
cycles

HLT Slots usage in LS2

- During the **LS2**, the Run 2 HLT farm was **intensively used** as opportunistic resource for offline processing.
- See CPU cores in use by CMS MC production and data processing jobs in **2019, 2020 and 2021**

30k CPU
cores



HLT Clouds for offline Computing in Run3

A **new farm** has been commissioned for **online** usage during the **Run 3**, i.e. run the trigger for the experiment:

Now, two cloud based HLT farms available in CMS

Run 2 HLT farm (aka P5 permanent cloud)

- **~25k CPU cores**, fully repurposed for offline computing
- Renamed to new CMS site name T2_CH_CERN_P5
- Hardware out of warranty
- To be decommissioned by the end of Run 3

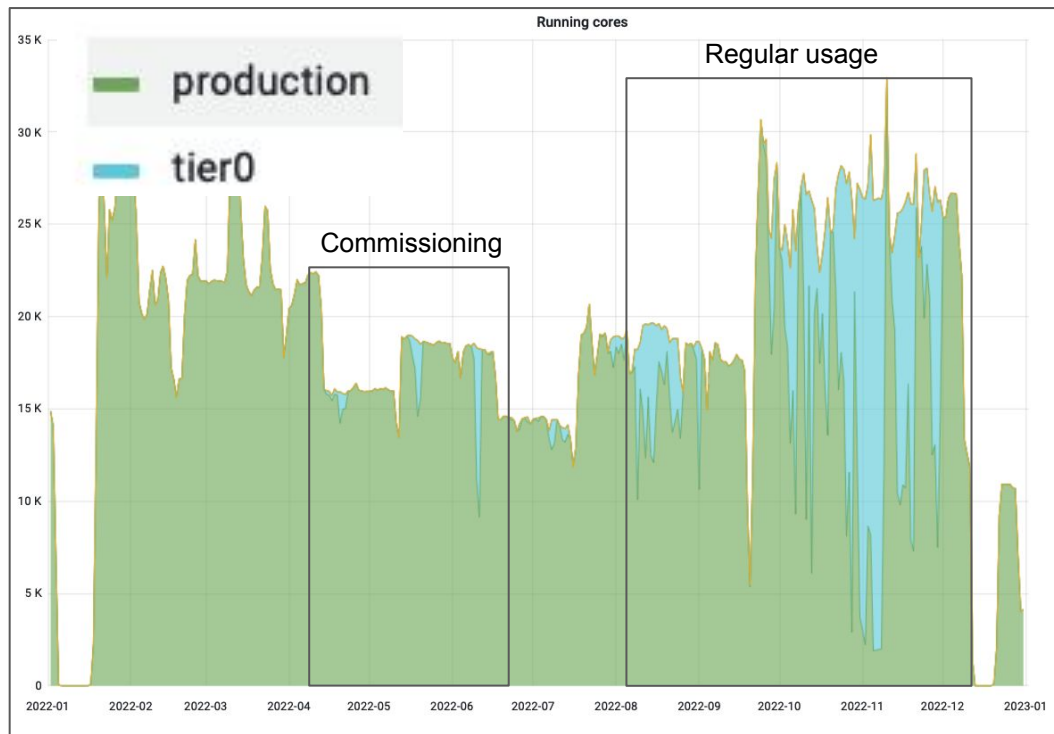
Run 3 HLT farm (aka Opportunistic cloud)

- Opportunistic cloud with **~ 30k CPU cores**
- Equipped with **dual NVidia T4 GPUs**
- Will only be used for offline computing during “no LHC data taking” period (e.g. YETS)
- T2_CH_CERN_HLT CMS site name now reserved for new HLT

Servers	CPUs	Disk (GB)	RAM (GB)	VMs/ Nodes	Total CPUs
Maguway	32	400	64	400	12800
Megware	24	300	56	219	5256
Action	28	400	56	280	7840
Total					~25K
Dell	128	-	256	260	~33K

Tier 0 jobs in the Run2 HLT farm in 2022

The Run 2 HLT farm (aka P5 permanent cloud) continued to be used opportunistically in the first year of the LHC Run 3, this time adding support for the Tier 0 in prompt data reconstruction tasks

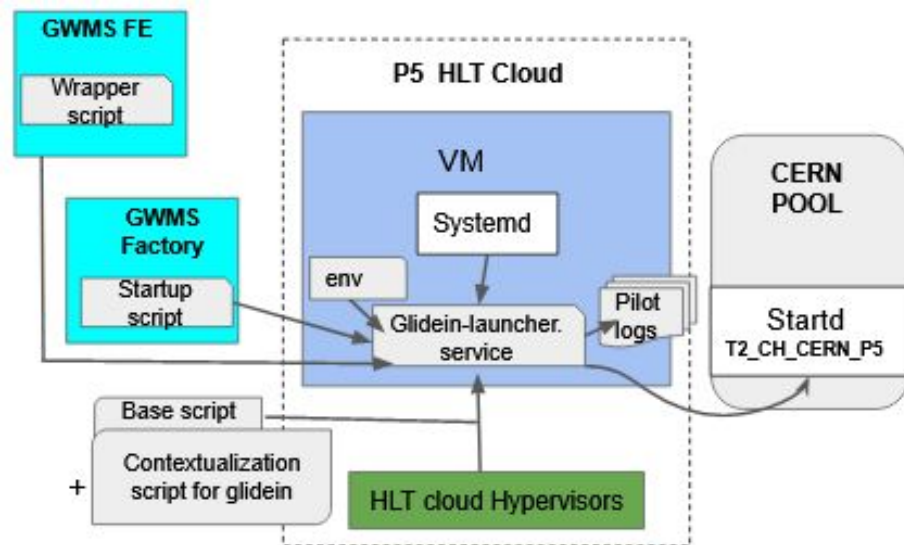


- **Commissioning** phase in Spring 2022 to ensure T0 tasks can reach old HLT farm and acquire slots at sufficient rate
- Once stable beams were declared, the resource was regularly **used as extension of the Tier 0 capacity**, as demanded by the data taking rates

Reimplementation of HLT with glideins in a Vacuum

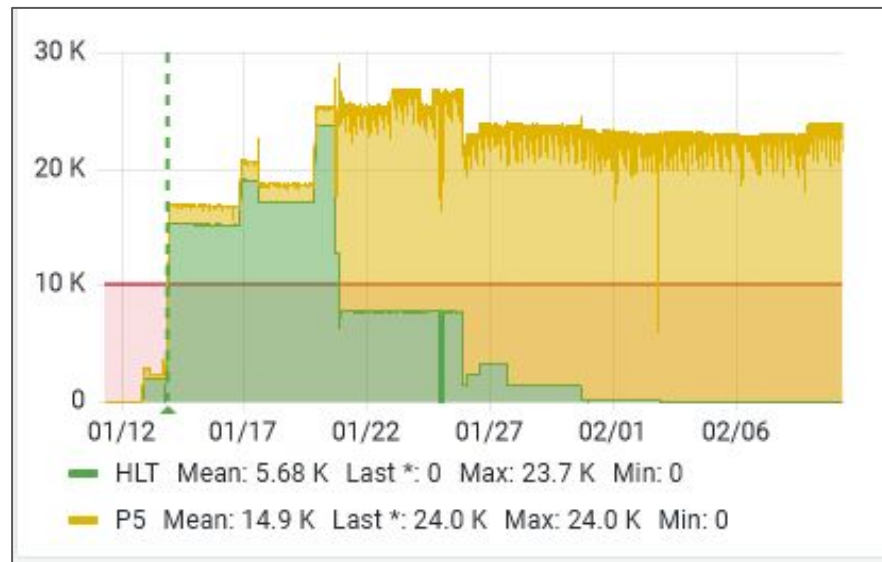
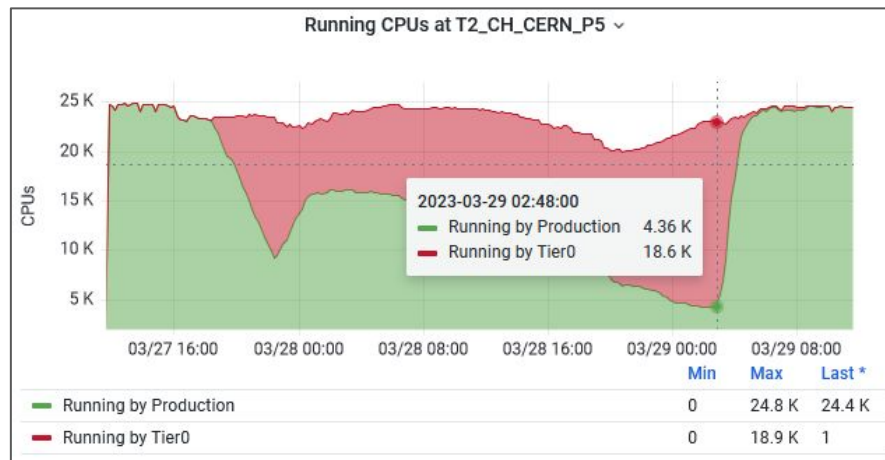
Switch from **static VM+startd** to a new configuration for the resource:

- Setting-up new site name “**T2_CH_CERN_P5**” for Glideins in vacuum implementation
- Glidein-launcher.service script is pushed to VM through contextualization script, without modifying original VM image
- Glidein-launcher.service, (systemd based service) pull the wrapper script from GWMS FE whenever it is updated, then launch glideins
- SI Generates and publish wrapper for T2_CH_CERN_P5 : Now configuration/tweaking and optimization of slots **can be done directly by the submission infrastructure team & relatively fast**
 - No need to coordinate with DAQ for draining pushing update/rebuild VM image



Reimplementation of HLT with glideins in a Vacuum

- The **transition** of CPU provisioning at the P5 permanent cloud into “glideins in a vacuum” mode started in **January 2023**
- Completely switched to **T2_CH_CERN_P5** by February



- **Commissioning** of the new site name and glidein mechanism with T0 jobs was then **successfully** tested, in preparation to the 2023 LHC data taking season

CPU efficiency and pilot overloading

Efficiency of Tier-1 CMS jobs in EGI accounting portal

Link [here](#)

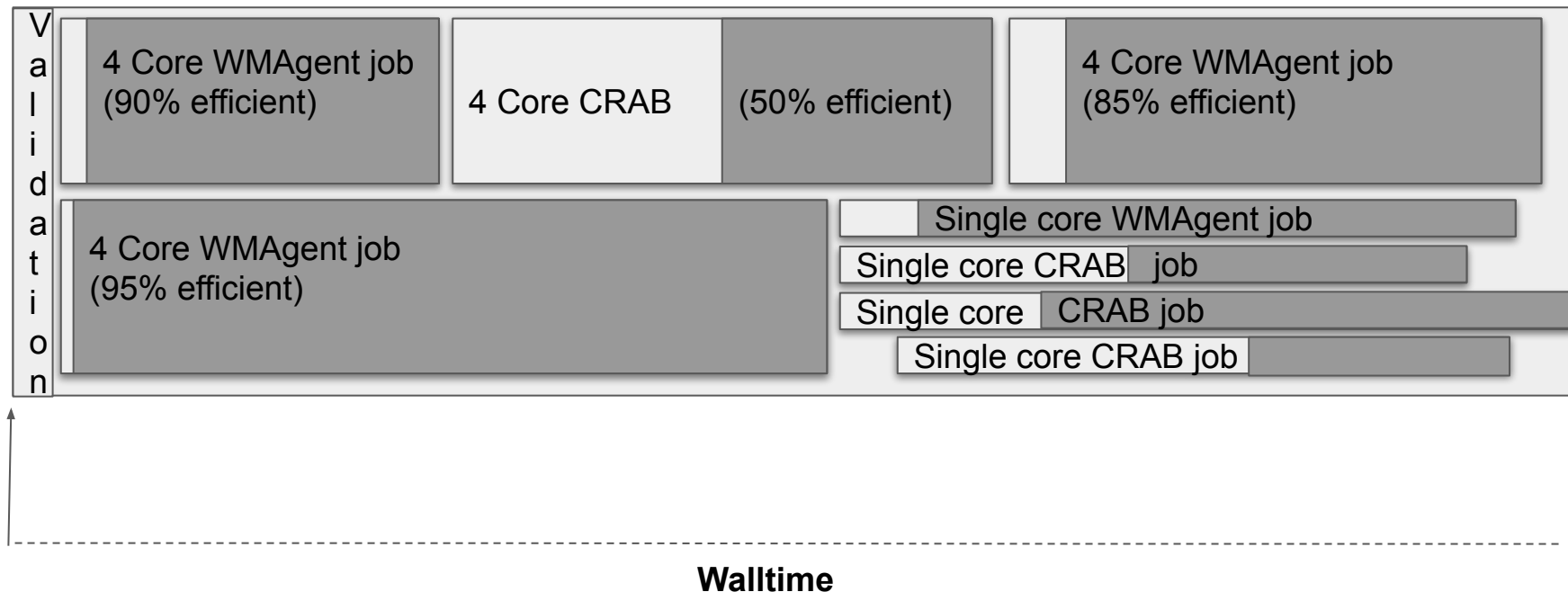
- EGI portal collects data reported by sites via the APEL tool
- The APEL parser processes logs of completed jobs
 - Creates one record per job
 - Sends records to EGI central server

Tier1 — CPU Efficiency (%) by Month and VO (LHC VOs)

Date	alice	atlas	cms	lhcb	Total
Sep 2022	80.81%	90.98%	75.22%	97.17%	84.8%
Oct 2022	70.71%	88.36%	79.74%	96.71%	82.94%
Nov 2022	70.7%	89.61%	82.02%	96.14%	83.78%
Dec 2022	82.42%	92.76%	83.03%	98%	88.37%
Jan 2023	71.64%	88.63%	79.42%	95.3%	83.32%
Feb 2023	53.35%	91.05%	75.06%	98.05%	79.94%
Mar 2023	77.35%	90.48%	79.09%	98.3%	85.85%
Apr 2023	76.59%	90.66%	77.75%	95.64%	85.5%
May 2023	69.12%	88.92%	76.24%	89.62%	81.54%
Jun 2023	57.71%	91.87%	80.11%	89.91%	82.17%
Jul 2023	50.01%	89.33%	82.7%	88.88%	80.72%
Aug 2023	50.97%	88.14%	80.98%	88.86%	79.81%
Sep 2023	46.76%	85.31%	72.48%	83.23%	74.1%
Total	67.52%	89.62%	78.86%	93.85%	82.6%

Efficiency calculated as “CPU Time / Walltime” (normalized)

A typical CMS “job”: 8-core 48h pilot job executing multiple payloads



WLCG Efficiency: “**CPU Time / Walltime**”

Legend
 **CPU Time**

Scheduling effects on CPU efficiency

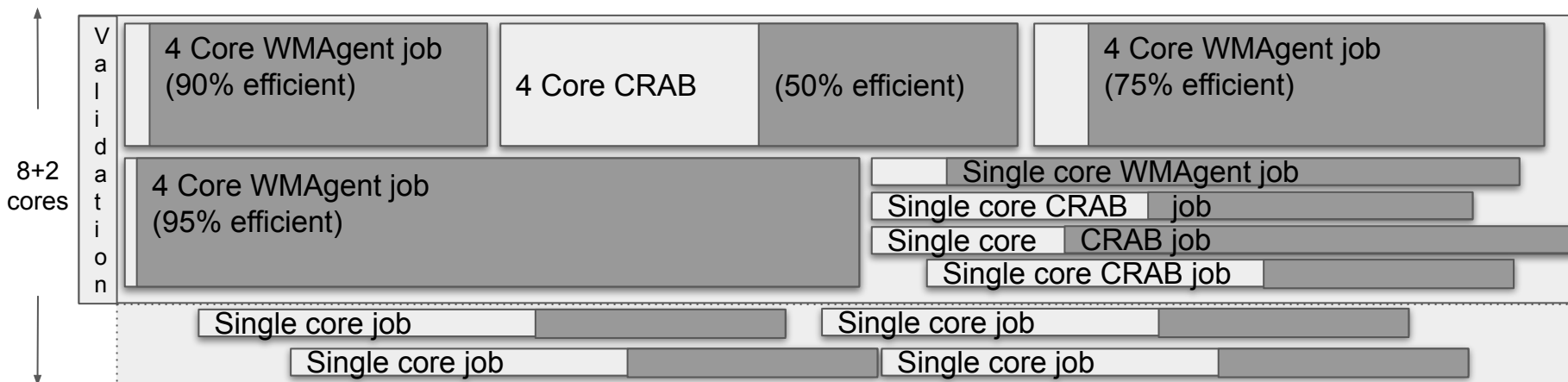
Scheduling effects on efficiency studied since the adoption of the multicore multi-payload pilot model (2016)

Inefficiency under control, at the order of 5%, classified to reasonable causes



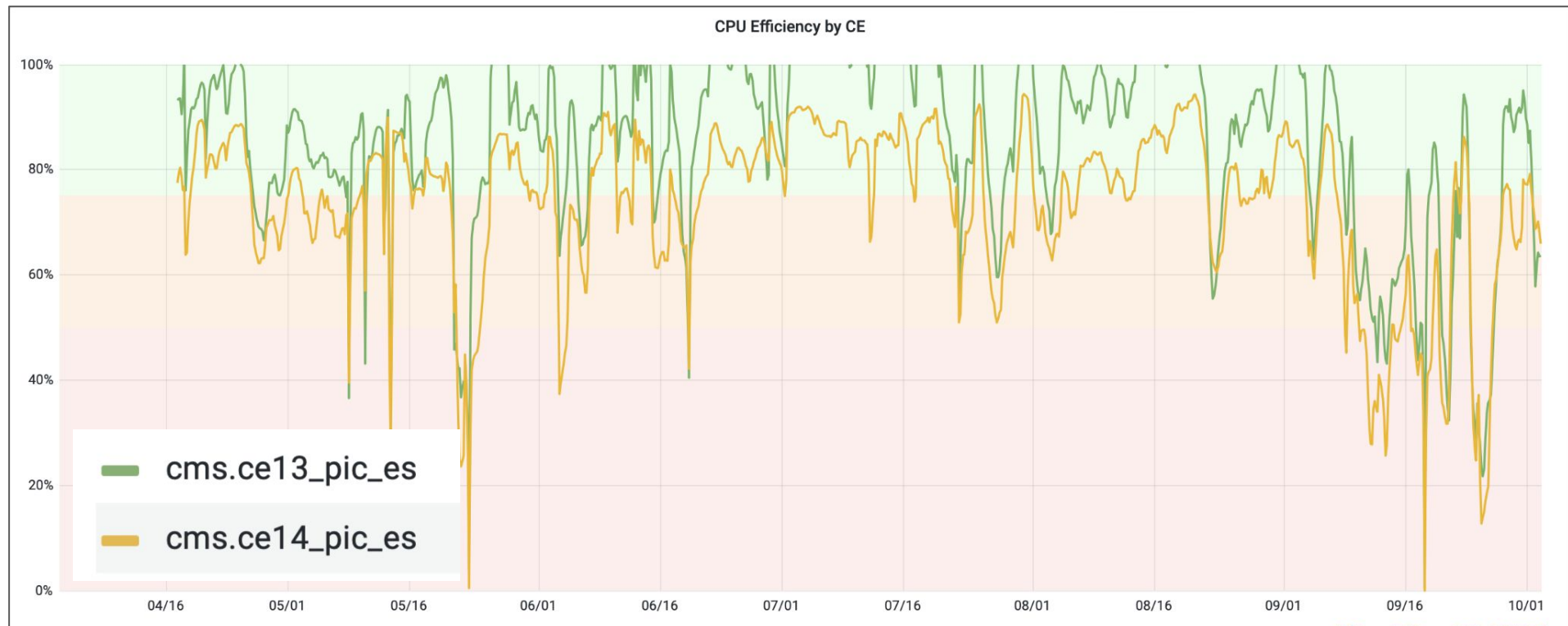
Q: Can we recover unused CPU cycles by pushing more work into the same pilot envelope?

- A simple idea is to “**enlarge**” our pilots so that **they accept more payload jobs into the same resources** acquired from the site
 - **Trivial to implement and test, so let’s try it!**
- **Example:** add 25% extra CPU cores and memory to the nominal values of our standard 8-core pilot
 - Provides 2 extra cores, e.g. available to run additional CRAB or production payloads



Overloading pilots tests

T1_ES_PIC case: CE13 has been receiving overloaded pilots for about 5 months



EGI efficiency of sites with overloaded CEs since January

Resource Centre PIC — CPU Efficiency (%) by Submit Host and Month (Custom VOs)

[Link](#)

Submit Host	Jan 2023	Feb 2023	Mar 2023	Apr 2023	May 2023	Jun 2023	Jul 2023	Aug 2023	Sep 2023	Total
ce13.pic.es:9619/ce13.pic.es-condor	71.58%	66.12%	77.66%	90.47%	86.59%	106.37%	95.59%	91.83%	70.2%	85.49%
ce14.pic.es:9619/ce14.pic.es-condor	71.58%	65.44%	75.97%	80.14%	74.64%	84.7%	80.64%	78.43%	56.26%	74.58%

Resource Centre CIEMAT-LCG2 — CPU Efficiency (%) by Submit Host and Month (Custom VOs)

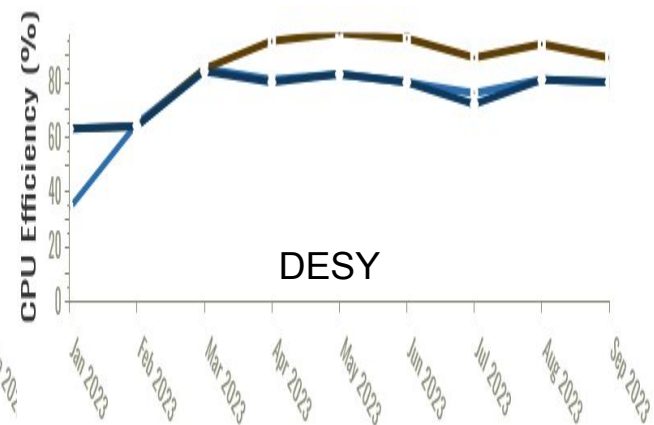
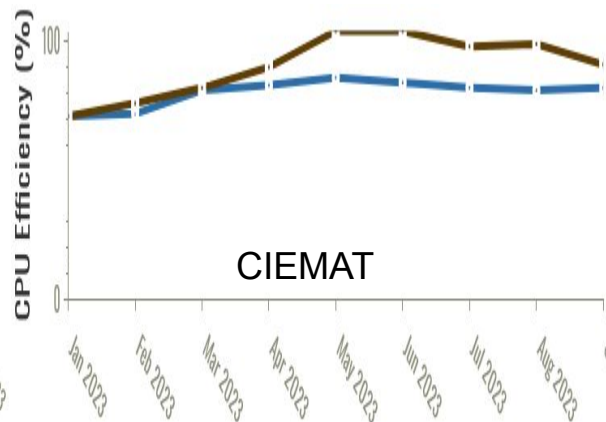
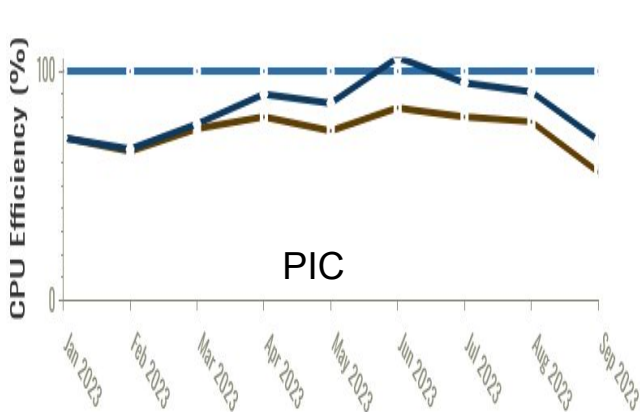
[Link](#)

Submit Host	Jan 2023	Feb 2023	Mar 2023	Apr 2023	May 2023	Jun 2023	Jul 2023	Aug 2023	Sep 2023	Total
condorce1.ciemat.es:9619/condorce1.ciemat.es-condor	71.06%	76.15%	82.41%	90.45%	104.08%	104.05%	98.85%	99.25%	91.38%	93.13%
condorce2.ciemat.es:9619/condorce2.ciemat.es-condor	71.01%	72.86%	81.36%	83.56%	86.01%	84.36%	82.8%	81.33%	82.15%	81.23%

Resource Centre DESY-HH — CPU Efficiency (%) by Submit Host and Month (Custom VOs)

[Link](#)

Submit Host	Jan 2023	Feb 2023	Mar 2023	Apr 2023	May 2023	Jun 2023	Jul 2023	Aug 2023	Sep 2023	Total
grid-htcondorce0.desy.de:9619/grid-htcondorce0.desy.de-condor	63.75%	64.6%	84.83%	80.32%	83.63%	80.76%	72.91%	81.03%	80.66%	76.89%
grid-htcondorce1.desy.de:9619/grid-htcondorce1.desy.de-condor	63.01%	64.44%	85.31%	95.24%	98.57%	96.76%	89.58%	94.53%	89.36%	87.2%
grid-htcondorce2.desy.de:9619/grid-htcondorce2.desy.de-condor	34.23%	65%	85.19%	81.33%	83.52%	80.73%	76.68%	81.9%	80.84%	79.65%



Conclusions

Conclusions and future work

- **The CMS SI team is in charge of the infrastructure where CMS processes data, including mission-critical T0 jobs**
- Incorporating new and more diverse resource providers
- **Periodically evaluates the scalability** of our infrastructure, in order to **anticipate** and **remedy** future scaling and stability problems
- **Resource heterogeneity** in HEP computing is a **key element** moving forward: CMS is adapting its WM and SI systems to properly manage new resource types and providers and put them to good use
- Achieving ever more sophisticated matchmaking strategies while maintaining scheduling efficiency
- Keeping up with changing technologies (so far tokens, ipv6, containers, EL9, etc)
- **Continuously supervise** the operations of the pool to ensure effective and efficient resource utilization in support of CMS research program